

Nikolai Madlener

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EDUCATION

Stanford University

March 2024 – September 2024

Master's Thesis and Visiting Researcher

Stanford, USA

- Building a hypertension medication recommendation system for remote patient management

Technical University of Munich

October 2021 – September 2024

Master of Science in Informatics (Computer Science)

Munich, Germany

- Focus on Software Engineering, Machine Learning, and Distributed Systems

Peking University

February 2023 – July 2023

General Visiting Student (Semester abroad)

Beijing, China

- Co-organized and spoke at PKU's first Exchange Cafe about LLM's and ChatGPT

Technical University of Munich

October 2017 – June 2021

Bachelor of Science in Informatics (Computer Science)

Munich, Germany

EXPERIENCE & INTERNSHIPS

Weavs GmbH

October 2023 – Present

Software Engineer (Working Student)

Dornbirn, Austria

- Development of Citymonitor, an IoT SaaS platform for smart cities using Elixir and the Phoenix framework

Technical University of Munich, Digital Health Group

October 2021 – March 2023

Software Engineer (Academic Assistant)

Munich, Germany

- Development of a navigation iOS app that improves users' fitness and health by supporting multimodal routing
- Improved average route search performance by over 50%

ALEAS AG

July 2019 – November 2022

Software Engineer (Working Student)

Ruggel, Liechtenstein / Hybrid

- Full stack web application development with .NET for internal and user-facing software with 2500+ active users
- Development of a cross-platform team-challenge app with React Native and the Firebase platform

Tesla, Inc.

March 2021 – May 2021

Logistics Software System Configuration (Internship)

Grünheide, Germany

- Initial configuration of the material flow software system for intralogistics at the Tesla Giga Berlin factory
- Requirements engineering and architectural conception based on exchange with material flow planners

PROJECTS

Stanford Spezi Framework Open-Source Contributions" | *Swift, Typescript, FHIR*

October 2023 – Present

- Implementation of the License and Spezi Rules Engine modules of the Stanford Spezi framework

Ferienakademie 2022 – "Decentralized Decision Making" | *Swift, SwiftUI, MQTT*

September 2022

- Research summer school in cooperation with Siemens for selected students aimed at exploring innovative ideas
- Development of a self-organized robot delivery system with focus on decentralized decision making

AWARDS & EXTRACURRICULAR EXPERIENCE

hackaTUM 2023 & 2022 Overall Winner

November 2022, 2023

- Overall winner of hackaTUM 2022 & 2023, one of EU's biggest hackathons, out of 220 teams (800+ participants)
- 1st place in the 'City of Munich' 2023 challenge (14 teams) and 2nd place in the 'Sixt' 2022 challenge (20 teams)

MINGA Mentor

August 2022 – February 2023

- Mentor for incoming exchange students to guide them during their time in Munich and at TUM

SKILLS

Languages: German, English (C1), Chinese (HSK1)

Programming Languages: Swift, Elixir, Typescript

Frameworks & Tools: SwiftUI, Phoenix, PostgreSQL, Firebase, Docker, Git, VS Code, XCode